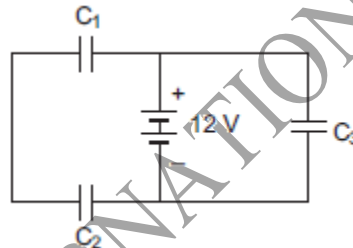
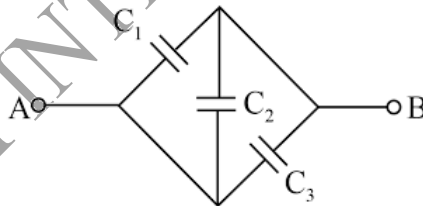


FREEDOM INTERNATIONAL SCHOOL
WORKSHEET - 2
PHYSICS
ELECTROSTATIC POTENTIAL & CAPACITANCE

1. (i) Two circular metal plates, each of radius 10 cm, are kept parallel to each other at a distance of 1 mm. What kind of capacitor do they make? Mention one application of this capacitor.
(ii) If the radius of each of the plates is increased by a factor of $\sqrt{2}$ and their distance of separation decreased to half of its initial value, calculate the ratio of the capacitance in the cases.
2. Three identical capacitors C_1 , C_2 and C_3 of capacitance $6\mu\text{F}$ each are connected to a 12 V battery as shown. Find
 - (i) charge on each capacitor.
 - (ii) equivalent capacitance of the network.
 - (iii) energy stored in the network of capacitors.

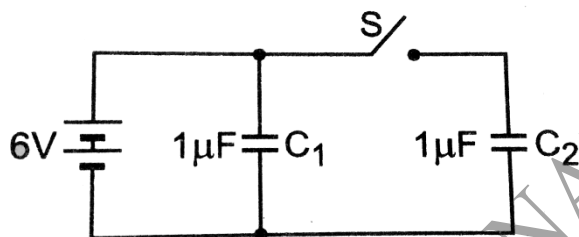


3. In Figure, $C_1 = 1\mu\text{F}$, $C_2 = 2\mu\text{F}$ and $C_3 = 3\mu\text{F}$. Find the equivalent capacitance between points A and B.

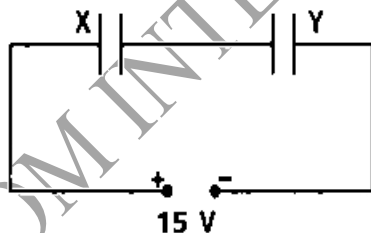


4. A parallel plate capacitor is charged to a potential difference V by a d.c. source. The capacitor is then disconnected from the source. If the distance between the plates is doubled, state with reason how the following will change: (i) electric field between the plates, (ii) capacitance and (iii) energy stored in the capacitor.
5. A parallel plate capacitor is charged by a battery. After some time, the battery is disconnected and a dielectric slab with its thickness equal to the plate separation is inserted between the plates. What change, if any takes place in (i) charge on the plates (ii) electric field intensity between the plates (iii) the capacitance of the capacitor, (iv) potential difference between the plates and (v) the energy stored in the capacitor? Justify your answer in each case.
6. A capacitor of unknown capacitance is connected across a battery of V volts. The charge stored in it is $360\mu\text{C}$. When potential across the capacitor is reduced by 120 V, the charge stored in it becomes $120\mu\text{C}$.

- (i) Calculate the potential V and the unknown capacitance C .
- (ii) What will the charge stored in the capacitor be, if the voltage applied had increased by 120 V?
7. Two capacitors of capacitances $3\ \mu\text{F}$ and $6\ \mu\text{F}$, are charged to potentials 2 V and 5 V respectively. These two charged capacitors are connected in series. Find the potential across each of the capacitors now.
8. The figure shows two identical capacitors, C_1 and C_2 , each of $1\ \mu\text{F}$ capacitance connected to a battery of 6 V. Initially switch 'S' is closed. After some time 'S' is left open and dielectric slabs of dielectric constant $K=3$ are inserted to fill completely the space between the plates of the two capacitors. How will the (i) charge and (ii) potential difference between the plates of capacitors be affected after the slabs are inserted?



9. Two parallel plate capacitors, X and Y, have the same area of plates and same separation between them. X has air between the plates while Y contains a dielectric medium of $\epsilon_r = 4$.
- (i) Calculate capacitance of each capacitor if equivalent capacitance of the combination is $4\ \mu\text{F}$.
- (ii) Calculate the potential difference between the plates of X and Y.
- (iii) Estimate the ratio of electrostatic energy stored in X and Y.



10. The figure shows a network of five capacitors connected to a 100 V supply. Calculate the total charge and energy stored in the network.

